

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING,
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Mid Semester Exam-I (July - Dec., 2025)

Course Name: Signals & Systems Course Code: MVPC-203 Max. Marks: 15 Max. Time: 50 Min.

Note: All questions are compulsory.

1. Given the signal $x(t)$ shown in Figure 1. Draw the signals $x(t+1)$, $x(-t+1)$, $x(\frac{3}{2}t+1)$ and $x(\frac{2}{3}t)$ with explanations. [5]

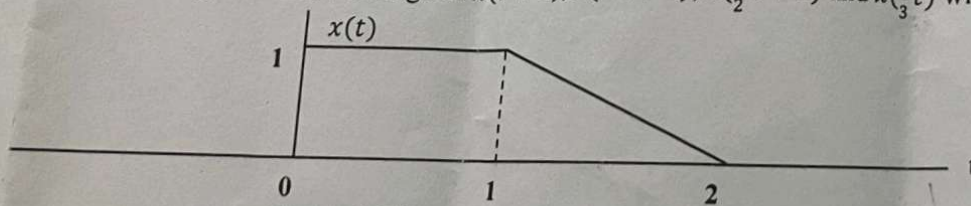


Figure 1. The signal $x(t)$

2. Find the Fourier series coefficients of $x(t) = 1 + \sin w_0 t + 2\cos w_0 t + \cos(2w_0 t + \frac{\pi}{4})$ and, also plot magnitude and phase of the Fourier coefficients. [5]
3. Find the Fourier transform of (a) $x(t) = e^{-a|t|}$, $a > 0$, (b) $x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases}$ → $\frac{\sin \omega T_1}{\omega}$ → Duality [5]